Edward P. Chandler

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Research Area

Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision

Education

University of Wisconsin–Madison Dec. 2027
Ph.D. in Electrical and Computer Engineering **GPA:** 4.00 / 4.00
- **Advisor:** Prof. Ulugbek S. Kamilov

Washington University in St. Louis
Transferred with Advisor after 2 years of Ph.D. research
- **Advisor:** Prof. Ulugbek S. Kamilov

Washington University in St. Louis May 2023
B.S. in Computer Science + Mathematics **GPA:** 4.00 / 4.00
- **Minor:** Philosophy-Neuroscience-Psychology
- **Research Advisor:** Prof. Ulugbek S. Kamilov
- **Academic Advisor:** Prof. Ron Cytron

Awards and Honors

NSF GRFP Honorable Mention	2024
McKelvey School of Engineering Valedictorian	2023
McKelvey School of Engineering Award for Research Excellence – CSE	2023
McKelvey Dean's Select Ph.D. Fellowship	2022

Experience

Computational Imaging Group, UW-Madison Madison, WI
Graduate Researcher Jan. 2026 – Present
- Design, analyze, and implement generative models (e.g. diffusion and flow-matching) to serve as data-driven priors for imaging inverse problems.
- Develop convex and non-convex optimization algorithms to solve high-dimensional inverse problems.
- Apply these frameworks to a variety of imaging modalities, particularly CT and MRI.
- Collaborate with other research labs on image reconstruction problems.
- Supervise undergraduate and Master students' research projects.

Los Alamos National Laboratory Los Alamos, NM
Graduate Researcher Jun. 2025 – Aug. 2025, May 2026 – Aug. 2026
- Designed and implemented large-scale Computed Tomography/Laminography (CT/CL) alignment and reconstruction algorithms.
- Developed proximal-based optimization algorithms.

Complex Media Optics Lab, ENS Paris Paris, France
Visiting Graduate Researcher Jan. 2024 – May 2024
- Implemented the system modeling and reconstruction algorithm for a novel Raman spectroscopy imaging system.
- Resulted in a publication in *Nature Communications*.

Computational Imaging Group, Washington University in St. Louis St. Louis, MO
Undergraduate Researcher Sep. 2020 – May 2023
Graduate Researcher May 2023 – Dec. 2025
- Designed, analyzed, and implemented Plug-and-Play image reconstruction algorithms.

- Analyzed the effect of distribution shifts on learned image reconstruction algorithms and designed algorithms to mitigate the effect.
- Trained denoisers and an image segmentation model on real data from WashU collaborators.

General Dynamics Information Technology St. Louis, MO
Computer Vision Data Science Intern Sep. 2021 – Aug. 2022
- Trained and optimized deep learning segmentation models for satellite imagery datasets.

The Trade Desk Boulder, CO
Software Engineer Intern Jan. 2019 – Apr. 2019
- Built a full-stack news app using The Trade Desk's database of internet users.

Teaching Experience

UW-Madison Madison, WI
Head Teaching Assistant, ECE 888: Topics in Mathematical Data Science Aug. 2026 – Dec. 2026
- Design assignments, give lectures, and conduct office hours.

Washington University in St. Louis St. Louis, MO
Teaching Assistant, ESE 351: Signals and Systems Jan. 2025 – May. 2025
- Designed assignments, give tutorials, and conduct office hours.

University of Bologna, Mathematics ML for Image Analysis Summer School Bologna, IT
Teaching Assistant, Optimization based machine learning for computational imaging Jun. 2024
- Designed and led tutorial on Plug-and-Play methods for image reconstruction.

Washington University in St. Louis St. Louis, MO
Teaching Assistant, CSE 247: Data Structures and Algorithms Sep. 2020 – Dec. 2020
- Led weekly group tutorials on analytical and coding problems.

Scholarly Activities

Invited Talks
- "Measurement Alignment for Computed Laminography," *Workshop on New Computational Methods for Imaging at Light and Electron Sources*, Bay Area Joint Light Sources Users' Meeting, 2026
- "Distribution Shift Estimation in Imaging Inverse Problems," *Special Session on Generative AI in Biomedical Imaging*, IEEE CAMSAP, 2025

Conferences Reviewed
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026
- European Conference on Computer Vision (ECCV) 2026
- Conference on Neural Information Processing Systems (NeurIPS) 2026

Journals Reviewed
- IEEE Transactions on Computational Imaging (TCI)
- IEEE Signal Processing Letters (SPL)
- SIAM Journal on Imaging Sciences (SIIMS)

Publications (* Equal contribution/Shared first authorship)

[1] S. Shoushtari, **E. P. Chandler**, and U. S. Kamilov. "Moments Matter: Posterior Recovery in Poisson Denoising via Log-Networks". In: *IEEE ICASSP*. 2026.

[2] A. Scott, E. Louwagie, **E. P. Chandler**, S. Kolluru, C. Fosher, A. Arter, F. Manyx, U. S. Kamilov, A. Odibo, and M. Oyen. "Predicting Stillbirth Risk Factors with Ultrasound Texture Analysis and Doppler Imaging of the Placenta". In: *Reproductive Sciences*, 2026.

[3] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov. "Closed-Form Approximation of the Total Variation Proximal Operator". In: *IEEE TCI*, 2025.

[4] S. Zhao, L. Chibani, **E. P. Chandler**, F. Liu, J. Hu, L. Valzania, U. S. Kamilov, and H. B. de Aguiar. "Computational field-resolved coherent chemical imaging". In: *Nature Communications*, 2025.

[5] W. Wang*, S. Shoushtari*, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov. "Distribution Shift Estimation in Imaging Inverse Problems". In: *IEEE CAMSAP*. 2025.

[6] S. Shoushtari, J. Liu, **E. P. Chandler**, M. S. Asif, and U. S. Kamilov. "Prior Mismatch and Adaptation in PnP-ADMM with a Nonconvex Convergence Analysis". In: *ICML*. 2024.

[7] **E. P. Chandler**, S. Shoushtari, J. Liu, M. S. Asif, and U. S. Kamilov. "Overcoming Distribution Shifts in Plug-and-Play Methods with Test-Time Training". In: *IEEE CAMSAP*. 2023.

Preprints (* Equal contribution/Shared first authorship)

[8] S. Shoushtari*, **E. P. Chandler***, X. Shi, and U. S. Kamilov. *SNaP: One-Step Posterior Sampling for Noisy Inverse Problems*. arXiv preprint arXiv:2609.34071. 2026.

[9] S. Shoushtari, **E. P. Chandler**, X. Shi, and U. S. Kamilov. *Fast and Faithful: Principled Conditional Flow Matching for Inverse Problems*. arXiv preprint arXiv:2609.12953. 2026.

[10] X. Shi*, **E. P. Chandler***, C. Y. Park, S. Shoushtari, and U. S. Kamilov. *NullFlow: One-Step Generative Reconstruction*. arXiv preprint arXiv:2606.22696. 2026.

[11] C. Y. Park, **E. P. Chandler**, Y. Hu, M. T. McCann, C. Garcia-Cardona, B. Wohlberg, and U. S. Kamilov. *Stochastic Generative Plug-and-Play Priors*. arXiv preprint arXiv:2604.03603. 2026.

[12] S. Shoushtari, **E. P. Chandler**, Y. Wang, M. S. Asif, and U. S. Kamilov. *Unsupervised Detection of Distribution Shift in Inverse Problems using Diffusion Models*. arXiv preprint arXiv:2505.11482. 2025.

[13] **E. P. Chandler**, S. Shoushtari, B. Wohlberg, and U. S. Kamilov. *Analysis Plug-and-Play Methods for Imaging Inverse Problems*. arXiv preprint arXiv:2509.15422. 2025.

Technical Skills

Programming & Tools: Python (PyTorch, NumPy), MATLAB, Linux/Bash, Git, Docker

Domain Knowledge: Computational Imaging, Deep Learning, Optimization, Inverse Problems, and Computer Vision